

Part A Chapter 8

From Cognition to Behaviour¹

1 Introduction

Behaviour of (human or other animal) organisms can occur in different types, varying from very simple behaviour to more sophisticated forms. Depending on the complexity of the behaviour of such an individual, viewed from an external observer's perspective, the internal cognitive representations and capabilities required to generate the behaviour also show a large variety in complexity.

2 Complexity of Environment, Behaviour and Cognition

From an evolutionary viewpoint, sometimes a relationship between the complexity of the environment and the complexity of mental representations and capabilities is assumed. For example, Peter Godfrey-Smith formulated the main theme of his book 'Complexity and the Function of Mind in Nature', as follows:

'The function of cognition is to enable the agent to deal with environmental complexity.'
(Godfrey-Smith, 1996), p. 3

In a more refined form, this *Environmental Complexity Thesis* can be formulated as follows:

- The more complex the environment, the more sophisticated the behaviour that is required to deal with this environment.
- The more sophisticated the behaviour that is required to deal with the environment, the more complex the cognitive representations and capabilities needed;

To deal with the physical environment, the evolution process has generated and still generates a variety of organisms that show new forms of behaviour (e.g., different from the behaviour of simple physical objects), by the development of cognitive capabilities of various degrees of complexity. The introduction of such organisms increases the complexity of the environment during the evolution process. Organisms that have to interact with these newly introduced

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organisms live in a more complex environment, and therefore need more complex (social) behaviour to deal with this environment, (to be) realised by more complex cognitive capabilities.

Given this global perspective, an important more detailed question remains which mental capabilities (and related representations), viewed from an internal cognitive perspective, are sufficient to generate which patterns of behaviour, viewed from an external behavioural perspective. This question is one of the two aims of this and the following chapters. This question is addressed by analysing cases of varying complexity.

3 External Behavioural Perspective and Internal Cognitive Perspective

For all patterns of behaviour discussed in subsequent chapters in Part B, a reflection will take place on questions such as:

- *From an external behavioural perspective:*
 - Which concepts relating to input and output of an individual are needed to describe the pattern of behaviour?
 - How can the dynamic relationships between input and output be described using these concepts?
- *From an internal cognitive perspective:*
 - What types of internal cognitive state properties are required to generate the behavioural pattern?
 - How can the dynamic relationships between these internal cognitive state properties be described?

These aspects will be used as a structuring principle in the next chapters; see also Table 1.

	concepts describing states	dynamic relationships describing processes
<i>external behavioural view</i>	<i>input and output state properties</i> - observed stimuli , incoming communication - actions, outgoing communication	<i>temporal relationships involving input and output state properties</i> - from observed stimuli and incoming communications to actions and outgoing communications
<i>internal cognitive view</i>	<i>cognitive state properties</i> - sensory representations, preparations - beliefs, desires, intentions - sensitivity levels	<i>temporal relationships involving cognitive state properties</i> - from sensory representations to preparations - from experiences to sensitivity levels - from sensitivity level to action - from desires to intentions - from intentions to actions

Table 1 The different perspectives used

4 Variations of Patterns of Behaviour and Cognitive Capabilities

The first case study (in the setting of a (pseudo-)experiment with animal behaviour) in Part B, Chapter 2 addresses the most simple behavioural pattern, *stimulus-response behaviour*: behaviour that solely depends on stimuli the individual gets as input at a preceding point in time. To describe such behaviour, usually an external perspective based on direct *functional associations* between input and output is taken, without any reference to internal or mental properties.

Following historical developments in the study of animal behaviour, a next class of patterns of behaviour, with slightly higher complexity, addressed (in the same experimental setting) is *delayed response behaviour*: behaviour that not only depends on the current stimuli, but also may depend on stimuli the individual had as input in the past; e.g., (Hunter, 1912; Tinklepaugh, 1932; Vauclair, 1996). Due to this temporal aspect, viewed from an external behavioural perspective, this pattern of behaviour cannot be described by direct functional associations between (current) input and output. The (external) description relating input states and output states over time, cannot make use of internal cognitive concepts; it necessarily needs a reference to inputs received in the past for this pattern of behaviour. Therefore, the type of external behavioural description becomes a temporal relationship, which is a bit more complex than in the previous case.

Viewed from an internal cognitive perspective, to describe mental capabilities generating such delayed response behaviour, often it is assumed that a memory in the form of some *internal model of the (current) world state* maintained by the individual is involved. Assuming such an internal world state model, a different description of the individual's process is possible. Within such an internal cognitive description, elements of this world state model play an intermediary *functional role*, mediating between inputs and outputs; e.g., (Kim, 1996; Maslin, 2001).

These two patterns of behaviour, stimulus-response and delayed-response, addressed in Part B, Chapter 2 can both be viewed as types of *reactive behaviour*. Other types of behaviour exist that go beyond reactive behaviour. For example, behaviour that does not depend on or depends in a more indirect manner on the individual's input in the present or in the past. Observed from the outside, this behaviour seems to come from within the individual itself, since no direct relation to inputs is recognized. It may look like the individual is in motion by reasons on its own, or motivated by itself to show a certain type of behaviour. To step beyond reactive behaviour, a third class of patterns of behaviour addressed in Part B, Chapter 3 is *motivation-based behaviour*.

An example pattern of externally observable (animal) behaviour is analysed which cannot be described as reactive behaviour. In this case the external behavioural description of the pattern of behaviour becomes still a bit more complex. To describe mental capabilities generating such a type of behaviour from an internal perspective, often the internal cognitive state properties *belief*, *desire* and *intention* are used. A desire can be just internally generated, but may depend on a history of inputs as well. Especially when desires dynamically depend on the history of the individual in a nontrivial manner, and not (only) on an assumed world state model within the individual, the individual's observed behaviour may be impressive and puzzling. Some not too complicated examples of such puzzles will be put forward and solved. In relation to this pattern of motivation-based behaviour, also the external attribution of beliefs, desires and intentions to other individuals is discussed, according to the so-called *intentional stance*; e.g., (Dennett, 1987).

A fourth class of patterns of behaviour, addressed in Part B, Chapter 4, which also goes beyond reactive behaviour, is *adaptive behaviour*. In this case, depending on its history the individual's pattern of externally observable behaviour is modified or tuned in a more or less sophisticated manner. An example pattern from this class is a simple form of *conditioning*: after being a victim of a stinging wasp, the individual behaves different in all future situations when a wasp comes close. Another example of adaptive behaviour addressed in Part B, Chapter 4 is based on the internal cognitive state property *trust*. An individual's behaviour in future situations is often modified depending on positive or negative experiences with earlier events in the individual's history. For example the individual will avoid as much as possible to buy fruit in a shop that earlier provided fruit of bad quality, due to which the individual lost trust in this shop. The internal cognitive state property trust is used to describe mental capabilities used to generate such a type of behaviour.

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